

Joydeep Das

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Professional Summary

GenAI & LLM Engineer specializing in RAG systems, multi-agent AI, and full-stack AI applications. Built MediRAG — a hallucination-resistant clinical AI system grounded in 1.1M+ medical records — and multiple production-grade AI systems spanning autonomous agents, health intelligence, and real-time research pipelines. Experienced with LangChain, LangServe, Ollama, Groq, and Google Gemini API. Passionate about building real-world AI products that scale.

Education

Amity University Kolkata, India

Bachelor of Technology – Computer Science and Engineering; GPA: 7.1/10 | July 2022 – Present

Unacco School Imphal West

Class XII, Percentage: 82% | 2022

Heritage Convent Imphal West

Class X, Percentage: 70% | 2020

Technical Skills

- **AI / LLM:** LangChain, RAG, Retrieval-Augmented Generation, Prompt Engineering, Multi-Agent Systems, LangServe, LangSmith, Ollama (Gemma, LLaMA 3), Groq, Hugging Face, Vector Databases, AI Agents, Generative AI
- **Backend:** Python, FastAPI, Node.js, Express.js, Flask, RESTful APIs
- **Frontend:** React.js, Next.js, Tailwind CSS, JavaScript (ES6+), HTML5, CSS3
- **ML / Data:** TensorFlow, Keras, Scikit-learn, Pandas, NumPy, BioBERT, FAISS, BM25
- **Tools & Infra:** Git, GitHub, Supabase, Vercel, Hugging Face Spaces, Make.com, n8n

Projects

MediRAG: Clinical-Grade AI Reasoning & Retrieval | repo | live

India Innovates 2026 National Hackathon | Rank: 62nd nationally | 2026

- Built a clinical AI system on 1.1M+ medical records with 107k+ indexed clinical vectors, reducing hallucinations via a 2-step dual-LLM verification pipeline with NLI-based faithfulness checks
- Hybrid retrieval engine combining BioBERT (semantic understanding) and BM25 (keyword matching) for high-precision medical Q&A
- Document-aware MediChat for PDF/report analysis with rich source citations, relevance scores, and NLI faithfulness checks
- Interactive 3D architectural visualization of the embedding and retrieval pipeline for clinical stakeholder clarity

FitnessBunny: CalPal Freemium | repo | live

GenAI + Full-Stack Health Intelligence Platform | Next.js 16, Supabase, Groq | 2026

- Replaced manual calorie logging with automated nutritional analysis using Groq-powered LLMs, improving tracking efficiency by ~60%
- AI-powered recipe generation engine transforming raw ingredients into personalized meal plans using structured AI pipelines
- Full-stack architecture with Next.js 16, Supabase (PostgreSQL), and NextAuth using a database-first RAG approach
- Premium UI with Tailwind CSS and Motion featuring glassmorphism, micro-interactions, and temporal calendar-based tracking

Researchix: Autonomous Research Agent System | repo

Distributed Multi-Agent AI + GenAI System | Groq, Tavily, arXiv, React | 2026

- 4-phase agentic pipeline (plan → retrieve → analyze → synthesize) processing academic content autonomously

- Real-time academic retrieval via Tavily Search and arXiv APIs with intelligent source filtering and structured data extraction
- High-speed LLM inference via Groq with NDJSON streaming for live agent reasoning and transparent workflow visualization
- Full-stack architecture: Express/Node.js backend on Hugging Face Spaces, React/Vite/Tailwind frontend on Vercel

Multi-Agent Bug Analysis System (MABAS) | repo

LLM-Powered Autonomous Debugging Pipeline | Groq + FastAPI + React | 2026

- 5-agent autonomous pipeline (triage → reproduce → analyze → patch → review) resolving bugs end-to-end with risk scoring (0–100)
- Dynamic bug reproduction via sandboxed Python execution generating real crash scenarios and stderr logs
- Groq (LLaMA 3.1 8B) for fast multi-agent reasoning; neo-brutalist React + Vite dashboard for live debugging visualization

Multi-Agent Launch Decision System (MLDS) | repo

Autonomous Multi-Agent AI Orchestrator | 2026

- Autonomous 6-agent system (Data, Marketing, Product, Engineering, Risk, Coordinator) generating real-time Proceed/Pause rollout decisions
- Deterministic JSON decision matrices with rationale, risk register, and communication strategy for full traceability
- Live React dashboard visualizing agent reasoning traces and tool outputs in real-time

Bacterial Species Classification System | repo

Deep Learning + Machine Learning Hybrid Model | 2025

- Bacterial classification model identifying 5 species with 95.83% test accuracy using VGG16 feature extraction + SVM classification
- PCA-based dimensionality reduction; CLI tool and Python API for single-image predictions and model verification

Additional projects (Vibe Check AI Weather App, Trending Movies Platform, Library Management API) available at github.com/JoyTheSloth

Achievements & Competitions

India Innovates 2026 – National Hackathon

Open Innovation Category | Rank: 62nd (national level) | 2026

- Built MediRAG — an AI healthcare system on 1.1M+ curated medical records — achieving national recognition in the Open Innovation category
- Deployed backend on Hugging Face Spaces for scalable model hosting; integrated FastAPI REST APIs for real-time inference

Work Experience

Lead UI/UX Designer

GEETBIH Labs Pvt. Ltd. | Part-time | April 2026 – Present

- Driving end-to-end design for digital products with a strong focus on usability, scalability, and impactful user experiences
- Leading creation of intuitive interfaces while building and maintaining a cohesive design system across web and mobile platforms
- Leveraging generative AI and prompt design to streamline workflows, enhance creativity, and accelerate the design process

App Designer

Modern Mahal | August 2025 – November 2025

- Designed intuitive, visually appealing web and mobile interfaces in Figma to align with brand identity
- Conducted usability reviews and implemented design refinements to improve navigation, engagement, and accessibility

Generative AI Developer Intern

AI Wallah | Remote | May 2025 – July 2025

- Developed and deployed LLM-powered web applications using Google Gemini API for Q&A and multi-turn conversational workflows
- Built full-stack integration of LLM services with Flask backend and JavaScript frontend through RESTful API architecture
- Accelerated AI feature development by ~50% through optimized prompt engineering and rapid prototyping

UI/UX Designer & Social Media Marketing

2GatherApp | Remote | February 2025 – October 2025

- Designed user-centric interfaces that enhance community engagement and streamline event discovery
- Built a consistent design system to unify app visuals and marketing assets
- Created and executed social media campaigns to boost reach, downloads, and brand identity

Web Designer

YGSD | September 2024 – November 2024

- *Project: Mourya Urja Matrimonial* — Designed user-friendly web layouts in Figma for a trusted matrimonial brand; integrated client feedback to enhance UX and accessibility
- *Project: Veliciae Jewellery* — Created visually cohesive web designs in Figma; collaborated with developers to translate requirements into effective design solutions

Certifications

Career Essentials in GitHub Professional Certificate

GitHub & LinkedIn Learning | 2025

Build Your Generative AI Productivity Skills

Microsoft & LinkedIn Learning | 2024

Career Essentials in Generative AI

Microsoft & LinkedIn Learning | 2024